KIIT University

School of Computer Engineering

SAP Business Data Cloud — Project Report

Record-to-Report (R2R)

Month-End / Year-End Financial Close

Fictitious Company: Kalinga Industries Ltd. (KAIL)

Bhubaneswar, Odisha | FY 2024-25

Submitted by:

[Student Name] | Roll No: [XXXXXXXX]

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1. Project Overview

The Record-to-Report (R2R) process is a core Finance function in SAP S/4HANA that covers the complete lifecycle from recording individual business transactions to producing audited, statutory financial statements. This project simulates the end-to-end R2R process for a fictitious Indian manufacturing company — Kalinga Industries Ltd. (Company Code: KAIL) — for Fiscal Year 2024-25.

The project leverages:

- SAP S/4HANA FI Module — General Ledger (GL), Accounts Payable (AP), Accounts Receivable (AR), and Asset Accounting (AA)
- SAP Business Data Cloud (BDC) — Financial data pipeline from S/4HANA to the analytics layer
- SAP Datasphere — Unified semantic data model for consolidated reporting
- SAP Analytics Cloud (SAC) — Period-end dashboards and variance analysis
- Custom ABAP Report — ZR2R_TRIAL_BALANCE for ALV-based trial balance output
- Python Automation — Period-end simulation, PDF generation, and Excel export

1.1 R2R Process Flow

The R2R cycle follows this sequence:

- Business Transaction Posting → Sub-Ledger Update → GL Account Update
- Period-End Closing Activities (Accruals, Depreciation, FX Revaluation, GR/IR Clearing)
- Trial Balance Extraction and Validation
- Posting Period Lock and Balance Carry Forward
- Financial Statement Generation and BDC Pipeline to SAC

2. Fictitious Company Setup — Kalinga Industries Ltd.

2.1 Company Master Data

Company Name	Kalinga Industries Ltd.
Company Code	KAIL
Country / Currency	India / INR (Indian Rupee)
Industry	Manufacturing — Specialty Chemicals
Fiscal Year	V3: April 1 to March 31
Chart of Accounts	KAIL
Controlling Area	KAIL
Registered Address	Plot 5, Industrial Area, Bhubaneswar, Odisha - 751010
Key Business Areas	BA01: Manufacturing | BA02: Trading
Cost Centers	CC_PROD (Production) | CC_ADMIN (Administration) | CC_SALE (Sales)
Plants	KL01 — Bhubaneswar Manufacturing Plant

2.2 SAP Organizational Structure

The company is configured in SAP with the following enterprise structure:

- Client: 100 (Production Client)
- Company Code: KAIL — Legal entity for all financial reporting
- Business Area: BA01 (Manufacturing), BA02 (Trading) — for segment reporting
- Controlling Area: KAIL — links FI to CO for cost management
- Plant: KL01 — production and inventory management location
- Purchasing Organization: KAPO — for MM purchasing
- Sales Organization: KASO — for SD order-to-cash

3. SAP Customization Steps (SPRO)

All configuration is performed via SPRO (SAP Project Reference Object) — the Implementation Guide (IMG). The following table lists all customization activities for the R2R setup:

T-Code	Activity	Configuration Detail
OX02	Create Company Code	Create KAIL — Kalinga Industries Ltd., Country IN, Currency INR
OB29	Fiscal Year Variant	Create V3: April-March with 12 posting + 4 special periods
OB37	Assign FY Variant	Assign V3 to Company Code KAIL
OBBO	Posting Period Variant	Create and assign posting period variant KAIL
OB52	Open/Close Periods	Configure which periods are open for posting
OB13	Chart of Accounts	Create KAIL chart of accounts, maintenance language EN
OB62	Assign CoA	Assign chart of accounts KAIL to company code KAIL
FS00	Create GL Accounts	Create all GL accounts per the chart of accounts structure
OBA7	Document Types	Define: SA (GL), KR (Vendor Inv), DR (Customer Inv), AA (Asset)
FBN1	Number Ranges	Define FI number ranges for company code KAIL, FY 2025
OBA4	Tolerance Groups	Set payment diff tolerance for GL/Vendor/Customer clearings
FTXP	Tax Codes	Create GST tax codes: V5 (5%), V1 (18%), A5 (5%), A1 (18%)
EC08	Copy Depreciation Chart	Copy reference chart 0IN (India) to create KAIL depreciation chart
OAOB	Assign Depr Chart	Assign KAIL depreciation chart to company code KAIL
AFAMA	Depreciation Keys	Define: LINR (Straight Line), DEGR (Declining Balance)

3.1 Key Configuration Notes

- Fiscal Year Variant V3 uses a non-calendar fiscal year (April-March), standard for Indian companies.
- Chart of Accounts KAIL is a custom chart created for this company; in real projects you may copy from INT (International).
- Tax codes follow Indian GST rates: 5%, 12%, 18%, and 28% slabs with both input (V) and output (A) codes.
- Depreciation chart 0IN (India) is copied to KAIL to ensure India-specific depreciation rules are applied.
- All number ranges use internal assignment with year-based ranges to avoid conflicts.

4. Pre-Close Trial Balance — FY 2024-25 (Period 12)

The following trial balance is extracted from SAP using T-Code F.08 or the custom report ZR2R_TRIAL_BALANCE. All amounts are in INR (Indian Rupees). The trial balance is BALANCED — total debits equal total credits at ₹ 5,02,00,000.

GL Account	Description	Group	Debit (INR)	Credit (INR)	Net Balance
100000	Cash and Cash Equivalents	Asset	₹ 45,00,000	—	₹ 45,00,000
100100	Accounts Receivable	Asset	₹ 87,50,000	—	₹ 87,50,000
100200	Inventory - Raw Materials	Asset	₹ 32,00,000	—	₹ 32,00,000
100300	Prepaid Expenses	Asset	₹ 2,50,000	—	₹ 2,50,000
110000	Plant & Machinery (Gross)	Asset	₹ 1,50,00,000	—	₹ 1,50,00,000
110100	Accumulated Depreciation	Asset	—	₹ 45,00,000	(₹ 45,00,000)
200000	Accounts Payable	Liability	—	₹ 52,00,000	(₹ 52,00,000)
200100	Short-Term Loans	Liability	—	₹ 30,00,000	(₹ 30,00,000)
200200	Accrued Expenses	Liability	—	₹ 7,50,000	(₹ 7,50,000)
200300	Tax Payable (GST)	Liability	—	₹ 4,50,000	(₹ 4,50,000)
300000	Share Capital	Equity	—	₹ 1,00,00,000	(₹ 1,00,00,000)
300100	Retained Earnings	Equity	—	₹ 38,00,000	(₹ 38,00,000)
400000	Revenue from Operations	Revenue	—	₹ 2,20,00,000	(₹ 2,20,00,000)
400100	Other Income	Revenue	—	₹ 5,00,000	(₹ 5,00,000)
500000	Cost of Goods Sold	Expense	₹ 1,32,00,000	—	₹ 1,32,00,000
500100	Employee Salaries & Wages	Expense	₹ 28,00,000	—	₹ 28,00,000
500200	Rent & Utilities	Expense	₹ 8,00,000	—	₹ 8,00,000
500300	Depreciation Expense	Expense	₹ 9,00,000	—	₹ 9,00,000
500400	Marketing & Advertising	Expense	₹ 3,50,000	—	₹ 3,50,000
500500	Finance Costs (Interest)	Expense	₹ 4,50,000	—	₹ 4,50,000
GRAND TOTAL			**₹ 5,02,00,000**	**₹ 5,02,00,000**	**BALANCED ✓**

✅ Trial Balance is BALANCED — Total Debit = Total Credit = ₹ 5,02,00,000

5. Month-End / Year-End Closing Checklist

The following activities are executed in sequence during the period-end financial close. All activities use standard SAP T-Codes and are tracked by the Finance team.

#	T-Code	Activity	Owner	Status
1	**FB50 / FBS1**	Post Accruals & Deferrals	Accountant	✅ **Done**
2	**AFAB**	Run Depreciation (Asset Accounting)	Asset Accountant	✅ **Done**
3	**FAGL_FC_VAL**	Foreign Currency Revaluation	Finance	✅ **Done**
4	**MR11**	GR/IR Account Clearing	AP Team	✅ **Done**
5	**FBICR1**	Intercompany Reconciliation	Finance	✅ **Done**
6	**F.08**	Run Standard Trial Balance	Controller	✅ **Done**
7	**ZR2R_TRIAL_BALANCE**	Custom ALV Trial Balance (ABAP)	Controller	✅ **Done**
8	**OB52**	Lock Posting Period	Finance Manager	✅ **Done**
9	**F.16**	Balance Carry Forward to New FY	Finance	✅ **Done**
10	**F.01**	Generate Financial Statements	CFO	✅ **Done**

5.1 Year-End Specific Steps

Step 1: [F.07] Carry Forward Open Items — Carry forward open AR and AP items to new fiscal year 2025-26.

Step 2: [F.16] GL Balance Carry Forward — Transfer retained earnings and carry forward GL balances to FY 2026.

Step 3: [OB52] Close FY 2025 — Lock all periods for FY 2025; open Period 1 for FY 2026.

Step 4: [F.01] Run Financial Statements — Generate Balance Sheet and Profit & Loss Statement for FY 2024-25.

Step 5: [BDC Pipeline] Trigger Year-End BDC Job — Execute full-year replication to Datasphere; publish SAC Board report.

6. Closing Journal Entries

The following journal entries are posted at year-end to ensure accurate financial statements. These are posted via T-Code FB50 (GL Document Entry) or automated via batch sessions.

#	Description	Date	Debit Account	Dr Amount	Credit Account	Cr Amount
1	Depreciation Accrual	31-Mar-25	500300 Dr	₹ 9,00,000	110100 Cr	₹ 9,00,000
2	Accrued Salaries	31-Mar-25	500100 Dr	₹ 2,50,000	200200 Cr	₹ 2,50,000
3	Deferred Revenue	31-Mar-25	400000 Dr	₹ 5,00,000	200200 Cr	₹ 5,00,000
4	Close P&L to RE	31-Mar-25	400000/400100 Dr	₹ 2,25,00,000	500xxx Cr + 300100	₹ 2,25,00,000

7. Custom ABAP Report — ZR2R_TRIAL_BALANCE

7.1 Report Overview

The custom ABAP report ZR2R_TRIAL_BALANCE is developed in SE38/SE80 to provide an ALV Grid-based trial balance specific to the R2R process. It extends the standard F.08 functionality with:

- Company Code and Fiscal Year / Period selection screen
- GL Account range and Account Type filters
- Debit / Credit split with net balance calculation from GLT0 table
- ALV Grid output with sort by Account Type and subtotals
- Double-click navigation to FS10N (GL Account Balance) for drill-down
- Top-of-page commentary showing company details and running totals

7.2 Key SAP Tables Used

Table	Name	Usage in ZR2R_TRIAL_BALANCE
GLT0	GL Account Period Balances	Main source for cumulative debit/credit balances per period
SKA1	GL Account Master (CoA)	Fetches account type (KOART), account group
SKAT	GL Account Short Text	Fetches account description (TXT50) in user language
T001	Company Code Master	Validates company code; gets chart of accounts (KTOPL)
BSEG	FI Document Segment	Used for drill-down line item analysis

7.3 How to Deploy in SAP

Step 1: [SE80] Open Object Navigator — Create a new program ZR2R_TRIAL_BALANCE under package ZFIN.

Step 2: [SE38] Paste ABAP Code — Copy the full ABAP code from src/abap/ZR2R_TRIAL_BALANCE.abap and paste into the editor.

Step 3: [Ctrl+F3] Activate — Activate the program — resolve any syntax errors for your system version.

Step 4: [F8] Execute — Run the program; enter KAIL, 2025, 12 on the selection screen.

Step 5: [ALV Grid] Review Output — The ALV grid shows trial balance sorted by account type with subtotals.

Step 6: [Double-Click] Drill Down — Double-click any line to navigate to FS10N for that account's balance detail.

8. SAP Business Data Cloud Integration

SAP Business Data Cloud (BDC) connects the R2R financial data in SAP S/4HANA to the SAP analytics ecosystem. This is the core of your course — the BDC layer enables unified, real-time financial reporting without manual data exports.

8.1 BDC Architecture for R2R

- S/4HANA FI — Source system for all journal entries, GL balances, and financial statements
- BDC Replication Pipeline — Automatically replicates CDS Views from S/4HANA to Datasphere
- SAP Datasphere — Semantic layer; hosts Business Entities, Analytic Models, and Fact Sheets
- SAP Analytics Cloud (SAC) — Consumes Datasphere models for dashboards and ad-hoc analysis

8.2 Key CDS Views for R2R

CDS View / Object	Purpose
C_GLACCTBAL_Q0001	GL Account Balance Query — aggregated by period, account, and company code
C_TRIALBCQ0001	Trial Balance Query — full debit/credit split view for all GL accounts
I_GLACCOUNT	GL Account Interface View — master data for accounts (joins SKA1 + SKAT)
C_FINSTATEMENT_Q0001	Financial Statement Query — Balance Sheet and P&L structure
I_JOURNALENTRY	Journal Entry Interface — individual posting lines for audit trail

8.3 SAC Dashboard Components

- Trial Balance Dashboard — Real-time balance view with period filter and account type breakdown
- P&L Variance Report — Actual vs. Budget comparison for revenue and expense accounts
- Cash Flow Monitor — Real-time cash position from GL account 100000
- Period-End Status Tracker — Checklist dashboard showing completion status of all closing activities
- Automated Alerts — Triggered when any account balance deviates more than 5% from plan

9. SQL Scripts — SAP HANA / Datasphere

The following SQL scripts are used to extract R2R data from SAP HANA (via SAP HANA Studio or DBeaver) or from SAP Datasphere. These scripts are included in the GitHub repository under src/sql/.

Script 1: Trial Balance Extraction (r2r_queries.sql — Query 1)

Extracts cumulative GL account balances from GLT0 for the selected company code and fiscal year. Calculates debit and credit splits and joins with SKAT for account descriptions.

Script 2: Accounts Receivable Aging (r2r_queries.sql — Query 2)

Queries BSID for open customer items and calculates aging buckets (Not Due / 1-30 / 31-60 / 61-90 / Over 90 days) based on due date vs. current date.

Script 3: GR/IR Reconciliation (r2r_queries.sql — Query 3)

Joins MSEG (goods receipt) with RSEG/RBKP (vendor invoice) to identify GR/IR mismatches that must be cleared before period close using T-Code MR11.

Script 4: Intercompany Balance Check (r2r_queries.sql — Query 4)

Aggregates BSEG intercompany postings by trading partner and GL account to identify unreconciled intercompany balances that must be resolved at month-end.

10. GitHub Repository & Version Control

10.1 Repository Structure

Repository URL: https://github.com/YOUR_USERNAME/kiit-sap-r2r-project

File / Folder	Contents
README.md	Complete project guide, process flow, setup instructions, and tool list
src/abap/ ZR2R_TRIAL_BALANCE.abap	Full ABAP source code for the custom trial balance ALV report
src/python/r2r_simulation.py	Python R2R simulation: trial balance, PDF report, Excel export
src/sql/r2r_queries.sql	SAP HANA / Datasphere SQL scripts for GL, AR, GR-IR, and IC queries
config/company_config.json	Company master data configuration in JSON format
data/sample/ trial_balance_sample.csv	Sample trial balance CSV for testing and reference
docs/R2R_Project_Report.docx	This Word document — full project report
docs/R2R_Project_Report.pdf	PDF version of the project report
.gitignore	Ignores: output/, __pycache__/, *.pyc, .DS_Store

10.2 How to Push to GitHub (Step-by-Step)

Step 1: [git init] Initialize Repository — Open Terminal / Command Prompt in the project folder. Run: git init

Step 2: [github.com] Create GitHub Repository — Go to github.com → New Repository → Name: kiit-sap-r2r-project → Create (do not initialize with README).

Step 3: [git remote add] Connect to GitHub — Run: git remote add origin https://github.com/YOUR_USERNAME/kiit-sap-r2r-project.git

Step 4: [git add .] Stage All Files — Run: git add . (stages all files; .gitignore excludes the output/ folder)

Step 5: [git commit] Commit with Message — Run: git commit -m "Initial commit: KIIT SAP R2R Project — FY 2024-25"

Step 6: [git push] Push to GitHub — Run: git push -u origin main (or master, depending on your Git version)

Step 7: [GitHub] Verify — Go to your GitHub repository URL — all files should be visible. Pin the repository to your profile.

10.3 Software Installation Guide

Tool / Software	Purpose	Version
SAP S/4HANA	Core ERP — FI Module (GL/AP/AR/AA)	S/4HANA 2023
SAP Business Data Cloud	Data pipeline & financial reporting cloud	BDC 2024
SAP Datasphere	Unified semantic data layer / warehouse	Cloud Edition
SAP Analytics Cloud (SAC)	Dashboards, variance & P&L analysis	Cloud Edition
SAP ABAP Workbench (SE80)	Custom report: ZR2R_TRIAL_BALANCE	NW 7.5+
SAP Fiori	Modern UX apps for period-end activities	Fiori 3.0
Python 3.11 + pandas	Simulation scripts & PDF generation	pandas 2.x
Git + GitHub	Version control & code repository	git 2.4+
Visual Studio Code	IDE for all code development	Latest

11. Conclusion

This project successfully demonstrates the complete Record-to-Report (R2R) process in SAP S/4HANA for a fictitious Indian manufacturing company — Kalinga Industries Ltd. (KAIL). The project covers:

- Enterprise structure setup including company code, fiscal year, chart of accounts, and document types in SAP.
- A complete, balanced trial balance for FY 2024-25 with 20 GL accounts across Assets, Liabilities, Equity, Revenue, and Expenses.
- All month-end and year-end closing activities with corresponding SAP T-Codes and ownership.
- Four critical year-end closing journal entries including depreciation accrual, salary accrual, deferred revenue, and P&L closure to Retained Earnings.
- A custom ABAP ALV report (ZR2R_TRIAL_BALANCE) that reads from GLT0 and provides a drill-down trial balance.
- SQL extraction scripts for SAP HANA / Datasphere covering trial balance, AR aging, GR/IR, and intercompany reconciliation.
- SAP Business Data Cloud integration design connecting S/4HANA FI data to SAC dashboards via Datasphere.
- A complete GitHub repository with all code, documentation, and sample data for version-controlled submission.

The R2R process is the backbone of financial reporting in any SAP-enabled organization. Mastery of this process — combined with SAP Business Data Cloud for modern analytics — provides the skills needed to operate as an SAP Finance functional consultant or a BDC data engineer in the industry.